

DO BLACK FINTECHS MATTER? THE LONG AND WINDING ROAD TO DEVELOP INCLUSIVE ALGORITHMS FOR SOCIAL JUSTICE¹

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Racism in the financial sector is a complex global phenomenon involving intertwined social, economic, and digital systems. Credit denial rates for Black applicants suggest that financial systems have assimilated racial discrimination into their algorithms and credit scoring tools. Considering that digital tools incorporate their developers' life experiences and perspectives, and in view of the low representation of the Black community in the digital startup universe, it is evident that even well-meaning fintechs are reproducing racial bias in their technology-intensive business models. In this paper, we investigate how three Black-owned Brazilian fintechs design and use inclusive algorithms to decrease persistent social injustice involving racial financial inclusion. We combine the concept of sociotechnical reconfiguration, taken from the South American tradition of "tecnologia social" (social technology), with the three core concepts of Nancy Fraser's theory of social justice—representation, recognition, and redistribution—to analyze the process by which Black-owned fintechs develop their particular solutions for combating racial bias. We found that pressured by the need to be profitable, the lack of capital to develop their own credit scoring tools, and the low profile maintained by the Black community ecosystem, Black-owned fintechs focus on a sort of affirmative experimentation to gradually create original digital solutions aimed at achieving social justice in the financial system.

Keywords: Algorithm racial bias, Black-owned fintechs, social justice, *tecnologia social* (social technology), financial inclusion

Introduction

Since the “Black Lives Matter” movement began to spread across the world over the last decade, particularly after the

brutal death of George Floyd, racial discrimination against Black communities has captured public attention in many different domains, including the financial sector. From the case of a former football player in the U.S.A. who was denied “private client” status at a big bank despite having millions of

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dollars in his account for years (Flitter, 2019) to the top executive of a Brazilian bank who said on TV that it was difficult to hire Black employees because the company would not “level down” by hiring unprepared workers (Santos, 2020), racial discrimination in the financial system has captured media headlines around the world. Apart from such mediatized cases, studies in the U.S.A. (Sanchez-Moyano & Shrimali, 2021), the U.K. (Adami, 2022), and Brazil (Paixão, 2017), demonstrate that Black communities are frequently underserved as customers within the financial system.

Similarly, the information systems (IS) literature shows that the emergence of the fintech movement over the last decade has fostered a belief in digital financial inclusion: empowering those who are financially underprivileged in the traditional financial industry (Lagna & Ravishankar, 2021). While some might believe in fintechs’ capacity to serve as the primary instrument for advancing financial inclusion for minorities, underserved populations continue to experience unequal access to financial services (Jones & Maynard, 2023). In fact, while fintechs may have the potential to expand financial inclusion, they have not yet significantly addressed the underlying causes of financial exclusion. As a result, even “well-meaning” fintechs can replicate inequalities in digital form, such as by charging higher interest rates for Black borrowers (Bartlett et al., 2020).

Two reasons might explain why fintechs might not be able to significantly reduce racial financial exclusion. The first is the fact that the Black community is underrepresented in the fintech industry (Carrilo et al., 2021). In the U.S. fintech industry, for example, only 2% of executives and 5.3% of professionals are African Americans (Brummer, 2020). Second, the limited representation of Black professionals in the financial sphere might be leading fintechs to reproduce the inequalities and biases of traditional credit scoring methods (Kane et al., 2021; Hollreiser, 2020). There is evidence that even in the absence of any direct information about race, credit score algorithms can incorporate racial bias (Vilarino & Vicente, 2020).

Yet despite the evidence that fintechs are lagging in ethnic diversity (Reynolds, 2020), academic research on fintechs led by Black entrepreneurs is difficult to find. A June 2022 search on Google Scholar for “Black fintech” and “Black-owned fintech” yielded no returns. Considering that the design of inclusive algorithms, an underinvestigated topic, is relevant to advancing social justice in our inequitable global society, we define socially inclusive algorithms as those designed to promote inclusiveness and diversity (De Cremer & De Schutter, 2021). We seek to understand the process by which Black fintechs create scoring mechanisms in the existing context of racially biased algorithms that reinforce oppressive social patterns and lead to financial exclusion. Therefore, we pose the following research question: *How are Black fintechs designing*

and using inclusive algorithms to decrease persistent racial social injustice regarding access to financial credit?

Our analysis of Black fintechs’ process of creating inclusive algorithms combines two theoretical streams: Nancy Fraser’s theory of social justice and the South American tradition of *tecnologia social* (social technology). From the theory of social justice (Fraser, 2009) we draw three interrelated core concepts—*representation, recognition, and redistribution*—encompassing symbolic meanings and rules operating under social structures that influence social interactions. From *tecnologia social*, we adopt the concept of *sociotechnical reconfiguration*, which focuses on the interplay between technical elements and social values involved in the design and use of a given *tecnologia social* (Dagnino et al., 2004; Pozzebon et al., 2021). Our empirical study examines three Brazilian fintechs founded and managed by Afro-Brazilian entrepreneurs (hereafter called Black fintechs) which created their own digital solutions for closing the financial inclusion gap that still throttles Brazil’s Black community. We sought to capture how Black fintechs are—or should be—developing inclusive algorithms to provide financial alternatives for the Black community.

Our study offers two main contributions. First, in terms of theory, we seek to enhance the role of IS research in promoting social justice through a better understanding of the design process and utilization of inclusive algorithms—particularly those dealing with credit scores, a crucial index of racial bias. By combining social justice theory with sociotechnical reconfiguration, we propose the theoretical concept of “affirmative experimentation” to explain the process of gradual development of digital solutions for combating racial discrimination and thereby advancing social justice in the financial sector. We believe that the concept of affirmative experimentation offers a particular contribution to the study of inclusive algorithms, a critical topic in the IS field.

Our second contribution is managerial in nature, bringing clearer understanding of the particular approach to design and utilization of inclusive algorithms that emerges from the affirmative experimentation concept. Our results suggest that in the development and use of inclusive algorithms, responsible financial institutions must pursue sociotechnical reconfiguration based on a combination of three elements: (1) socially excluded people must themselves take the lead in processes aimed at their inclusion (representation, in Fraser’s terms), (2) socially excluded people must embrace their own identities in order to transform the symbolic order embedded in the exclusionary reality (Fraser’s recognition), and (3) scoring development and use must combine human with automated analysis to be effective in fighting exclusion (Fraser’s redistribution). Although these elements may be present in the literature on avoiding minority and algorithm bias, combining

the three in a perspective of affirmative experimentation represents an original contribution to the orientation and training of responsible managers and developers with an interest in resisting social exclusion in digital systems.

This introduction is followed by our theoretical background, with an overview of the literature on racial financial inclusion and fintechs. We then discuss how the racial bias embedded in algorithms is carried over to financial scoring systems, closing with a summary of gaps identified throughout our literature review. Next, we present our theoretical framework and methodological design, followed by sections for results and discussion. We conclude with final considerations, indicating the main contributions of our research as well as some of its limitations.

Theoretical Background

The Persistent Racial Financial Exclusion

Financial inclusion is a widely recognized instrument for driving development, helping people escape poverty, and managing financial emergencies that can push families into destitution (Demirgüç-Kunt et al., 2020). Although numerous studies have examined ways of promoting financial inclusion for a low-income population, scholars have recently begun to acknowledge the need for distinctive approaches when addressing the financial needs and challenges facing Black communities (Sanchez-Moyano & Shrimali, 2021).

Black people have undergone a long history of exclusion from traditional financial systems. For example, U.S. banks reject Black people's loan applications more than twice as often as those of White applicants (Mikhitarian, 2018). Similarly, in Brazil, a study of over a thousand micro-entrepreneurs suggests that loans requested by Afro-descendent entrepreneurs were less likely to be approved than those requested by White entrepreneurs. Additionally, even if approved, Black borrowers received smaller loan amounts at higher interest rates, a phenomenon attributable to the presence of discriminatory mechanisms such as stereotyping and prejudice (Paixão, 2017).

In view of the limited progress achieved through public policies in reducing the racial wealth gap (Baradaran, 2018), Black communities have created their own alternatives to deal with financial exclusion. Symptomatic of this search for expanded financial access, the creation of Black-owned banks has long been observed, particularly in the U.S. (Ammons, 1996). Nonetheless, Black-owned banks account for only 0.5% of all community banks in the U.S. despite their ability to support Black communities and do what nonminority banks don't do,

such as tolerating higher risk and increasing services during crises (Neal & Walsh, 2020).

By focusing on the financial needs of Black communities, Black-owned banks operate *perforce* in less affluent—hence, less profitable—areas, rendering them less profitable than nonminority banks of similar size (Kashian et al., 2014). This operational condition limits their ability to accumulate sufficient capital to fully respond to Black communities' financial needs (Chiong et al., 2018). This explains why, in and of themselves, Black-owned banks might not be able to provide a solution to racial inequalities—a solution essential to the growth of the overall economy (Sanchez-Moyano & Shrimali, 2021).

Florant et al., (2020) identified certain elements that might reduce racial bias in credit offering to Black families. First, it is necessary to connect Black families with financial institutions close to where they live. Second, there is a need for greater diversity in the financial system, in which Black communities are usually underrepresented. Third, the use of digital financial platforms should be promoted by exploring innovative and inclusive credit decision-making—combining big data, artificial intelligence, and behavioral economics—as an alternative to conventional backward-looking credit scoring focused on previous history.

Financial Inclusion of Black Communities and Fintechs

The emergence of the “fintech revolution” over the last decade appears to have addressed the underlying causes of persistent inequality in the financial system. In fact, the literature on fintechs tends to associate their emergence with expectations that racial barriers in traditional loan distribution channels can be overcome (Hauptert, 2022). These studies link fintechs with the democratization of the financial system (Lagna & Ravishankar, 2021), financial inclusion (Senyo et al., 2021) and, indirectly, greater social justice in the financial sector (Tantri, 2021). Other scholars have suggested that although fintechs enable unbanked consumers to access digital financial services, they do not necessarily reduce financial exclusion based on race (Hayashi & Ruiz, 2021). Kane et al., (2021) provided one of the reasons for such apparent discrepancies: fintech designers may adopt algorithms that, for example, use zip codes and geographic information on historically segregated neighborhoods. As a result, predictive models built around this type of information may incorporate race-based discrimination (Hollreiser, 2020).

Just as the Black community is underrepresented in the tech industry's overall workforce (Roberts & Mayo, 2019), the fintech industry is also characterized by an extremely limited

presence of Black entrepreneurs (Brummer, 2020; Carrilo et al., 2021). This scenario is repeated across many other fields, rendering the Black community “underrepresented, tokenized, isolated, marginalised and excluded from positions of power” (Payton, 2021, p. 222). Since algorithms are created by people and thus incorporate human choices that represent the life experiences of their creators and reproduce their implicit biases (Martin, 2019), the low representation of Black professionals in the financial sphere (Reynolds, 2020) might be a factor leading fintechs to reproduce the racial inequalities and biases of traditional scoring methods.

Following the 2008 global financial crisis, a new wave of fintechs emerged, driven by a social entrepreneurial logic aimed at leveraging entrepreneurship to address social problems (Langley, 2020; Siqueira et al., 2023). While social entrepreneurship offers the potential for a more equitable future, the way they make sense of social problems and develop solutions (Kimmitt & Muñoz, 2018) is not immune to the influence of racial structures. Like any other type of organization, they may either enhance or diminish the agency of racial groups (Ray, 2019). For instance, Almandoz (2012) demonstrated that the particular background of a local bank’s founder and the bank’s embeddedness in specific situations impact the firm’s institutional logic. Similarly, designers’ situational positioning and contextual factors of the design process can influence the materiality of the digital artifacts they create (Gero & Kannengiesser, 2007).

Thus, decreasing the impact of restrictive lending practices requires ensuring that minorities take on a protagonist role in financial institutions (Eyal-Cohen, 2021). This usually involves founding their own, a goal reflected in recent efforts to support Black banking institutions (Gantt, 2021). The Black money movement, originating in the U.S. and emerging more recently in Brazil, seeks to reduce racial inequalities by promoting access by Black-owned businesses to financial services provided by the Black community itself (Soares, 2021). This movement recognizes the need to go beyond acquiring economic capital for Black businesses by also fostering human capital (through training and education) and social capital (supporting networks and ecosystems) as a way to strengthen Black entrepreneurship, especially in periods of crisis (Barki et al., 2021). Being based in entrepreneurial ecosystems (Alaassar et al., 2022), fintechs were targeted by the Black money movement, which openly promoted the establishment of Black-owned fintechs. In this view, Black-owned fintechs would appropriate the technological tools that currently reinforce racism in the financial system and, instead, make them function to eliminate racial bias. In other words, the building of financially inclusive algorithms for the Black community would occur concomitantly with more Black-led fintechs (Allocca, 2021).

Algorithm Racial Bias and Credit Scoring Systems

There are social consequences of algorithmically based decision-making processes. Despite a certain misperception of the neutrality of algorithms, they can, in fact, reinforce oppressive social constructions for certain groups (Silva, 2020). Problems related to algorithm bias are not new, and the racial issue is well captured in the literature—in particular, the literature pertaining to credit access (Chander, 2017). Efforts to prevent algorithmic technologies from reinforcing and reproducing oppressive social constructs (Benjamin, 2019) necessarily involve specifying the objectives and instruments available to develop the system. Investigating racial bias in health care systems, Obermeyer et al. (2019) suggested that algorithmic bias results from the selection of convenient variables that seem effective but, in fact, merely reflect structural inequalities. This is exemplified by choosing cost-related variables to predict patient health needs rather than illness-related variables. These authors consider the judicious choice of data labels to be a fundamental requisite for reaping the benefits of algorithms while minimizing their risks in socially relevant areas. This means that, from the outset, the system’s general assumptions must be carefully defined in order to mitigate any reinforcement of social oppression of minority groups.

Van den Broek et al. (2021) proposed a “hybrid” approach to minimize bias in machine learning algorithms. Based on a study of hiring systems, they found that combining algorithms and human experts in a mutual learning process blends experts’ situated knowledge with knowledge generated by machines in the hiring process. This human-machine hybrid, combining algorithms with practice-based knowledge, is difficult to achieve if the developers are disconnected from the domains in which their tools operate. This means intense interaction between developers and substantive experts is required to provide the continuous feedback necessary for improving algorithm development.

Also, carefully choosing the variables to be included in the algorithmic model and constant interaction between human experts and algorithm developers are critical to avoiding the biases embedded in mechanisms of structural oppression (Noble, 2018). Thus, filtering out racial bias in algorithms will depend on human experts applying their life experience to grasp the significance of those variables potentially subject to structural exclusion and maintaining a continuous iterative relationship between algorithm developers and human experts’ knowledge to preclude the reinforcement of racial stereotypes (Sweeney, 2013). Moreover, there is a need for further studies investigating how human intervention can prevent or reduce the impacts of algorithmic bias (Kordzadeh & Ghasemaghahi, 2021) and unintended negative

consequences. A prominent example is the widespread discrimination that accompanies transferring decision-making agency from humans to algorithms across a wide range of possibly applicable areas (Rinta-Kahila et al., 2021).

In addition to the literature on algorithmic racial bias, we also reviewed studies on credit scoring systems. Credit scoring is a particular type of lending analysis algorithm designed to assess the financial capacity of a person or household. The process of assigning a single quantitative measure—or score—to a potential borrower estimates the borrower's future loan performance while simultaneously lowering the loan transaction costs and improving the accuracy of credit evaluations for lenders (Robb & Robinson, 2018).

Credit scoring relies mostly on an individual's financial information—including their bank account profile, outstanding debts, financial statements, bill payment history, income, employment history, and property—as well as on personal information, such as their profession and permanent residence, among other indices. For business borrowers, information considered in credit scoring also includes the particular industry, physical location, employment, sales, profits, and financial capital.

However, even in the absence of any direct information about race, credit score algorithms can incorporate racial bias. A study sponsored by a large fintech company shows that Black people receive lower credit scores, making them more likely to be underbanked (Broady & Hester, 2021). This study, aimed at providing recommendations for “responsible fintechs” to bridge the racial financial gap, suggests the modernization of credit-scoring mechanisms through the adoption of machine learning technologies and alternative data sources to replace the traditional backward-looking approach. While well-intentioned, these suggestions may not achieve the expected racial gap-closing results in the financial world. Vilarino and Vicente (2020) showed that even in the absence of any direct information about race, credit scoring that is driven by machine learning incorporates racial bias based on the borrower's location information. In the same way, alternative data can perpetuate the discriminatory cycle if used to target vulnerable populations in promoting high-interest loans (Jones & Maynard, 2023). This happens, first, because racial bias is present in many types of algorithms and can be incorporated into credit decisions that serve only to reinforce discrimination in the loan process (O'Neil, 2016). A second factor is that data collected from mobile, public geospatial, and satellite sources have the potential to harm certain minority groups (Simumba et al., 2022).

Another approach is to shift from traditional backward-looking credit scoring mechanisms, which assess a borrower's situation

retrospectively, to a forward-looking scorecard which focuses on measuring a customer's current repayment potential. Indeed, Robb and Robinson (2018) found that such forward-looking measures of creditworthiness result in similar predictions for White and minority business borrowers in terms of their future ability to pay, thus indicating a lesser prevalence of racial bias with this approach. However, the forward-looking approach could also be developed into some type of psychometric or social algorithm that is not without risk to minorities. Siqueira et al. (2023) explained how forward-looking credit scoring models have profound implications for the economic autonomy of micro-entrepreneurs. These models involve continual data extraction and monitoring, effectively enforcing surveillance mechanisms that come to dictate the economic behavior of loan applicants.

The combination of human and machine knowledge, also known as hybrid intelligence (van der Aalst, 2021) or supervised machine learning (Lehr & Ohm, 2017), appears to be instrumental in countering racial bias in credit scoring. Van der Aalst (2021) maintains that humans outperform machine decision-making in areas that require contextual knowledge, adaptability, and empathy—particularly in boundary cases of credit scoring where racial bias may come into play. Lehr and Ohm (2017) point to credit scoring as likely suitable for supervised machine learning, given unsupervised algorithms' potential for discrimination against minorities.

Gaps Identified from the Literature Review

Although the insights provided by the extant literature increase our knowledge of racial financial exclusion, further research is needed to elucidate *how* to increase social justice by addressing algorithmic racial bias and improving credit scoring systems. Our research points out the need for the IS field to target the development of inclusive credit scoring algorithms that expand financial services to marginalized communities. While credit scoring is a common subject in the IS literature, discussion of how to build inclusive credit scoring is less prevalent (Wang et al., 2019). Similarly, while studies of algorithmic bias, particularly those related to race, have taken on added significance (Obermeyer et al., 2019), only a small portion of them target credit exclusion (Vilarino & Vicente, 2020). The rare critical literature dealing with credit exclusion based on algorithmic bias focuses on explaining the causes of the problem (Robb & Robinson, 2018; Fuster et al., 2022) or simply listing measures to minimize it (Broady & Hester, 2021). Theorizing concerning solutions to solve it is rare (Majchrzak et al., 2016). This scarcity opens the door to a theoretical contribution that offers IS researchers new perspectives from which to explore the role of algorithms, with a view towards decreasing/closing the credit market's racial exclusion gap.

Theoretical Positioning

The theory of social justice developed by Nancy Fraser (2009) over the last 30 years has not often been mobilized by the IS community; a few exceptions are Marjanovic et al. (2022), Peña Gangadharan and Nikklas (2019), and Krishna (2021). Our proposed theoretical framework combines Fraser's theory of social justice with the Latin American concept of sociotechnical reconfiguration (Dagnino, 2019) as a theoretical lens to clarify the potential relationship between fintechs, inclusive algorithms, and race.

Fraser's approach to social justice presents three interconnected core concepts that are useful in considering racial financial inclusion: *redistribution*, *recognition*, and *representation*. These three concepts represent symbolic meanings and rules operating under social structures that shape social interactions. Fraser (1995, 2000) had previously conceptualized *redistribution*, which corresponds to the economic structure of society, and *recognition*, relating to culture.

Redistribution includes property systems, labor division, and social classes—all distinguished by their differing potential to allocate resources. Therefore, remedies for economic injustice encompass some kind of political-economic restructuring that potentially involves income redistribution, reorganization of the division of labor, submission of investment decisions to democratic control (affirmative approach), or, more radically, transformation of economic structures (transformative approach) (Andersen et al., 2009).

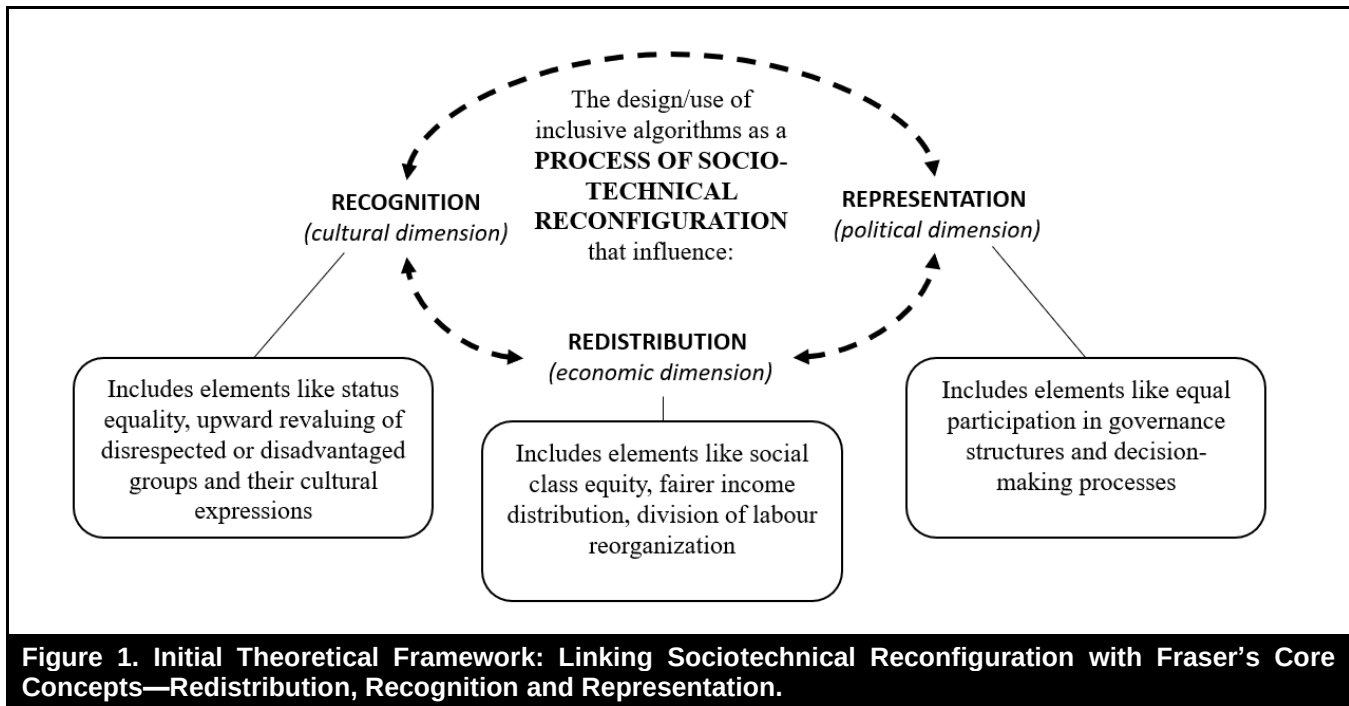
Recognition involves the status order of society, wherein a particular status group's socially entrenched patterns of cultural value are distinguished by the relative honor, prestige, and esteem they enjoy vis-à-vis those of other groups (Andersen et al., 2009). Fraser denoted two approaches to carrying out redistribution and recognition: affirmative and transformative (Fraser et al., 2004). Affirmative recognition seeks to impart value to devalued identities, while transformative recognition seeks to transform the symbolic order (Fraser et al., 2004). Remedies for social injustice entail some type of cultural or symbolic change, involving upward revaluation of the disrespected identities and cultural expressions of the disadvantaged, recognition and positive valuation of cultural diversity (affirmative approach), or, more radically, transformation of societal patterns of interpretation and communication (transformative approach).

More recently, Fraser added the third conceptual element, *representation*, which introduces a more explicitly political dimension to the previously identified economic and cultural components of social justice. Taken together, redistribution, recognition, and representation allow for the problematization of governance structures and decision-making procedures. Representation concerns political participation, or the

opportunity to take part in decision-making, which is central to the conceptualization of economic needs, making it a pivotal element in leveraging social responses to those needs. Significantly, representation extends beyond the judicial or political sphere and implies taking part in decisions involving a wide range of situations affecting individuals and groups, both in the workplace and in the boardroom (Burton, 2014).

The second theoretical influence on our theoretical framework comes from southern countries. *Tecnologia social* (we avoid the use of "social technology" in English because it has a different meaning than the original term in Portuguese/Spanish) represents a solid theoretical tradition which has developed over the course of a century (1) starting in India with Gandhi's movement against British imperialist domination, (2) becoming a source of inspiration for the appropriated technologies stream in the 1970s, and (3) being consolidated in South America in the 1990s (Pozzebon & Fontenelle, 2018). One of the main struggles in its historical development has been its persistent political criticism of the universalist view of Western-based science and technology development imposed across all continents (Dagnino, 2019). *Tecnologia social* offers a sharp contrast, implying the design, adaptation, and use of technologies in which the often excluded or discriminated periphery are central protagonists. *Tecnologia social* thus represents a decolonial theoretical stream that problematizes the center/periphery boundary, not only in terms of the international division of labor but also with regard to hegemony in technology development (Thomas & Bunch, 2013; Smith et al., 2014). It is worth emphasizing that the term technology is applied in a broad sense—including both hard and soft, ranging from material devices to immaterial methodologies, and even encompassing ways of organizing. *Tecnologia social* therefore implies the design/adaptation and use of material and immaterial technologies where the local community is the central protagonist, giving primacy to local needs and skills without neglecting global/technical knowledge and assistance (Saldanha et al., 2022).

Among the several ideas that make up the *tecnologia social* research stream, our work focuses on one central concept: *sociotechnical reconfiguration*. It helps to capture the interplay between technical elements and social values inscribed in a changing process, and it was inspired by a mix of northern and southern theoretical and practical influences, including Feenberg's (2002) critical theory of technology, and, more recently, solidarity technoscience (Dagnino, 2019). This is an ongoing process in which social practices, cultural frames, and material artifacts are mutually reconfigured with a view towards technological appropriation by minorities. Such a process considers social and technical dimensions as inseparable, therefore bearing a certain resemblance to recent sociomateriality debates. Figure 1 summarizes our theoretical framework combining sociotechnical reconfiguration with the three elements of Fraser's social justice theory.



Methodological Design

Doing Collective Case Studies: Following Three Brazilian Black Fintechs

To address our research question—*How are Black fintechs designing and using inclusive algorithms to decrease persistent racial social injustice regarding access to financial credit?*—we developed a methodological design that respects Stake's (1998) definition of an instrumental and a collective case study. Instrumental case studies are chosen not for the intrinsic interest of the case itself but for their suitability in exploring research questions that transcend the case, thus bringing a plausible understanding to similar cases. This understanding is a step towards extending the analytical generalization from few to many cases (Stake, 1998). When we select more than one instrumental case, we have a collective design.

We applied purposeful sampling (Patton, 1990) criteria to select three Brazilian Black fintechs, which are called Fintechs A, B, and C for reasons of confidentiality. We define *Black fintechs* as financial technology startups that are founded and managed by Afro-Brazilian entrepreneurs, with a specific focus on serving the Black community. Providing digital banking services such as digital accounts, payment services, and credit granting, the three Black fintechs in our study are at the early stages of development. Because of their

rapid growth, they have drawn growing attention from the national media for their innovative and inclusive purposes. However, regarding financial *credit scoring* services, they are at different stages of maturity and have different scales, providing us with a richer understanding of the process of developing inclusive credit scoring algorithms. The selected fintechs are particularly relevant cases as they provide a timely opportunity to develop new theories on the design and implementation of inclusive algorithms in the financial credit domain. By pursuing analytical generalization (Stake, 1998), which differs from generalization of a statistical nature, the knowledge produced from careful analysis of these three Brazilian cases can shed light on other cases of financial inclusion in similar or comparable contexts.

Data Collection and Analysis

Our research team has been conducting research on financial inclusion and fintechs for several years. In this context, between 2018 and 2019, we made our first contact with Fintech A. In the following years, we observed the emergence of other Black fintechs, including Fintechs B and C, stimulating our intellectual curiosity and leading us to more intense interactions with members of the Black community. Ultimately, we initiated a research project on Black fintechs. To obtain a deep understanding of this emerging phenomenon, we conducted 24 semi-structured interviews from July 2021 to June 2022. Among these, two respondents were interviewed twice, and on two occasions,

interviews were conducted with two participants simultaneously. Each session lasted an average of 60 minutes. The interviews were recorded on audio and video and transcribed. We triangulated the rich empirical material gathered from interviews with documents (website content, news, reports, articles), videos, field notes from observations, and informal conversations with actors in the fintechs' and Black communities' ecosystems.

The interviews followed guidelines based on our theoretical framework. Our questions sought to grasp (1) the sociotechnical reconfiguration process characterizing each fintech, the role of Black people in the governance and business model design, their interaction with and use of algorithms, and (2) the role of the three central concepts of social justice theory: redistribution, recognition, and representation.

We began by conducting interviews with the founders of each Black fintech, followed by interviews with managers and technical staff members. Additional interviews were then conducted with actors in the Black fintechs' ecosystem. All respondents in the study identified themselves as Afro-Brazilians, except one. They ranged in age from late 20s to

early 40s, with 45% being female. The majority of participants had successful careers as executives or entrepreneurs in various industries, leaning heavily towards technology and finance. A detailed description of the respondents and the role of the interviews for the data analysis is described in Table 1.

Although the number of interviewees for each Black fintech may seem relatively small, we interviewed the totality of founders, executives, and staff members working in customer relations and credit scoring—which, in some cases, represented almost the entirety of the fintech's employees. (Having a small number of full-time employees is typical of early-stage startups in general. Fintech B, for example, is essentially made up of the two founders, who are responsible for the entire process of management and credit analysis—with support activities being outsourced.)

Our study opted for a phenomenon-focused problematization approach to engage with theory building based on new empirical phenomena (Monteiro et al., 2022). The data analysis was driven by an interactive and abductive process (see Figure 2), honed by our critical view (Alvesson & Kärreman, 2011; Locke et al., 2022).

Table 1. Summary of the Interviews

Data source	Role for the analysis
<i>Semi-structured interviews with members of Fintechs A, B, and C:</i>	
Fintech A Founder 1A (female) Founder 2A (male) Manager 1A (male) (2 interviews) Manager 2A (female) Developer 1A (male) Fintech B Founder 1B (male) (2 interviews) Founder 2B (female) Fintech C Founder 1C (male) Manager 1C (male) Service-provider 1C (female) Service-provider 2C (female)	The interviews with the founders were essential to understanding their history, vision, and the business/operational model of each fintech. The interviews with other actors (managers and technical staff) provided valuable insights into the transformations, demands, barriers, and challenges faced by Black fintechs.
<i>Semi-structured interviews with actors of the Black fintechs' ecosystem:</i>	
• 4 social entrepreneurs (all female) of non-profit organizations^a • 5 Target clients (2 male and 3 female) of Black businesses^b • 3 data analysts (all male) of other fintechs • 1 academic/expert (male) on Brazilian Black context	These interviews were important to draw a rich picture of the Black fintechs' ecosystems and contexts. They offered important insights into the needs and barriers faced by Black communities and about the role of Black fintechs in financial inclusion.
<i>Total 24 interviews (average 60 min each)</i>	

Note: ^aTwo social entrepreneurs from the same organization were interviewed together, as per their request. ^bTwo target clients from the same businesses were interviewed together, as per their request.

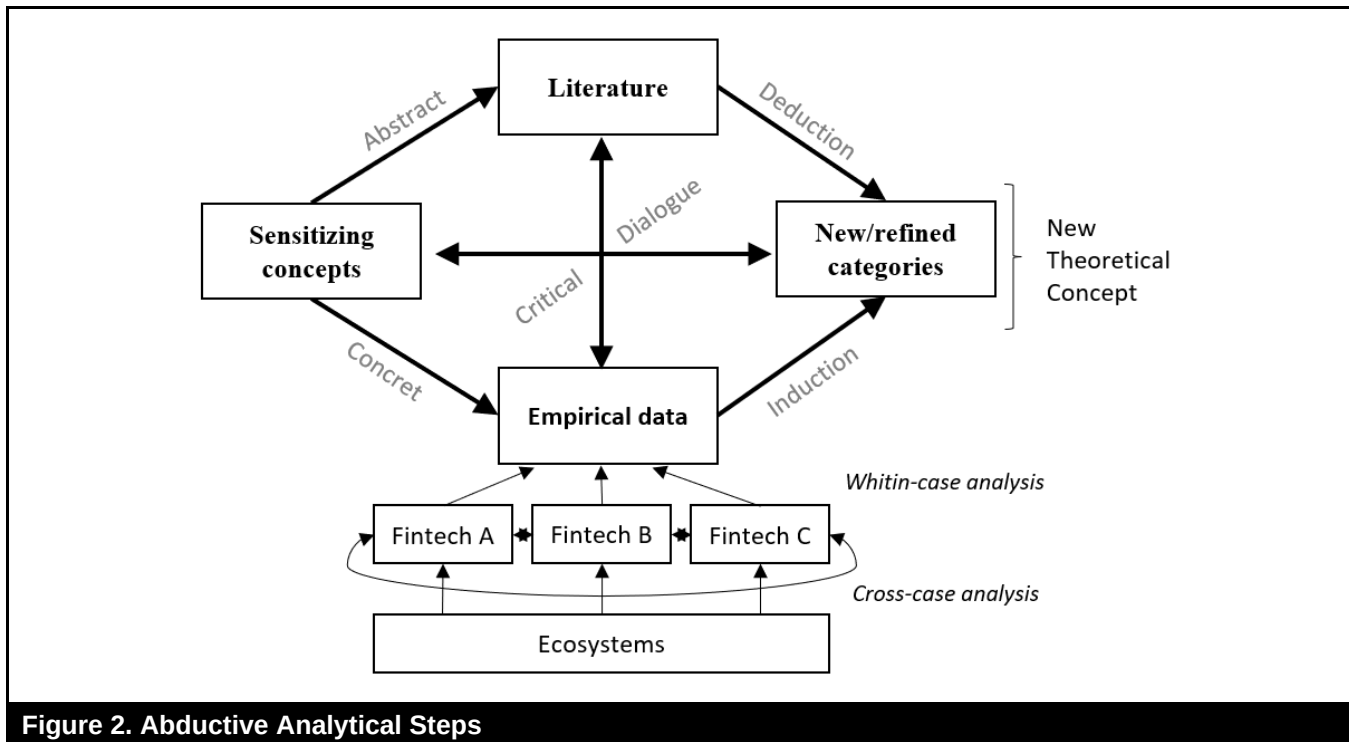


Figure 2. Abductive Analytical Steps

Table 2. Summary of the Data Structure

Dimensions	Categories	Strategies
Recognition	Race/class recognition	Fostering Black individual self-esteem
		Creating Black community identity
		Recognizing race as signifier of class
		Recognizing peripherals as market potential
Representation	Black people's protagonism	Increasing Black protagonism within fintechs
		Increasing Black protagonism in the financial ecosystem
Redistribution	Credit access	Running forward-looking analysis
		Testing affirmative variables
		Building contextually tailored algorithms
		Integrating racial bias awareness

The first reading of the empirical material brought deep understanding of our three cases. Then, the theoretical lens was mobilized to guide multiple cycles of transcript reading, particularly the interviews with the Black fintech's founders. Our sensitizing concepts (i.e., redistribution, recognition, and representation) served to identify convergences and disparities across the three cases. As certain "mysteries" came up (Alvesson & Kärreman, 2011), additional interviews with actors of the ecosystem were conducted to clarify and challenge our understanding of the emerging phenomenon of Black fintechs. Throughout this process, categories and strategies emerged, as illustrated by Table 2.

The abductive process of data analysis was further supported by descriptive memos and external documents, such as news clips and founder interviews given to third parties, which served to complement and enrich our understanding. Data were coded using both digital (Word and Excel) and manual (post-it notes) tools; in-vivo codes were outlined to sustain the identification of the new categories and strategies, which were interactively discussed and reevaluated by the research team.

Findings

Our findings are organized into four subsections. The first describes our cases, i.e., the three Black fintechs. The second relates to the contextual background that triggered the emergence of those Black fintechs in Brazil. The third presents the result of the application of the theoretical framework: the identification of categories and strategies that show *how* these Black fintech are operating to decrease racial injustice in financial credit. The concluding fourth subsection presents our final theoretical framework.

Presenting the Three Brazilian Black Fintechs

Fintechs A, B, and C are three distinct companies, founded and managed by Black entrepreneurs in Brazil, that share a common goal of promoting financial inclusion among marginalized populations, particularly among Black individuals and small businesses. The founders and executives of the firms, which have enjoyed high rates of growth, are highly educated Black entrepreneurs who share similar histories and the social purpose of including Black and poor people in the financial system.

Fintech A was founded by two Black entrepreneurs to address the limited access to credit faced by Black individuals and small businesses who have been excluded by traditional banks. It pursues this objective by offering digital wallets, insurance, and microcredit. Currently, Fintech A has over 18,000 customers, with 57% of them being Black, and plans to extend its reach to 100,000 accounts in the near future. The founding motivation behind the company's establishment was the personal experience of one of its founders, who faced challenges in accessing credit (Corrales & Diniz, 2020). Fintech A began its credit operation with small-scale experiments to provide loans to Black entrepreneurs who were rejected by conventional banks. While this approach succeeded in redistributing credit to some who had been previously excluded, Fintech A took advantage of Brazilian regulations (Diniz et al., 2014) to partner with a specialized credit scoring institution. This required a return to traditional credit scoring models, a shift that reduced Fintech A's ability to approve loans to Black individuals. Nonetheless, it has continued to interact with its clients to collect new data and encourage positive financial actions.

Founded by two young Black entrepreneurs with backgrounds in economics and finance, Fintech B is a digital bank that transforms small businesses into a network of "correspondent banks" (see Leonardi et al., 2016, for more on this financial model). With over 110 units and 30,000 clients, the firm uses a proprietary digital platform to provide services that include bill payments, withdrawals, and microcredit. Its business model focuses on offering credit to marginalized populations who

have been excluded from traditional financial systems, including people with criminal records, unpaid debts, or no financial history. However, this redistributive mission doesn't override the importance of maintaining a profitable portfolio. As a result, only 20% of Fintech B's total portfolio goes to allocation of credit to the hitherto economically excluded.

Fintech C captured growing attention due to its innovative approach to financial inclusion. Founded just before the COVID-19 pandemic, the company initially offered digital payment services but quickly developed services to manage charity donations from large industries to support those impacted by the crisis. The company has since grown to serve over 30,000 customers. Fintech C was founded by a Black entrepreneur with a commitment to providing marginalized populations with financial services. By using race and socioeconomic status as positive criteria in its allocation process, the company has been able to provide Black individuals with credit cards and donated grants.

A Context Triggering the Emergence of Black Fintechs

Brazil's Afro-Brazilian population is among the largest outside Africa, constituting over half of the country's overall population. However, institutionalized racism persists in numerous sectors of Brazilian society, including education, employment, and the criminal justice system (Nascimento, 2016). Data from the country's official statistics for 2021 indicated that 56.1% of the Brazilian population identifies as Black, with 47% identifying as light-skinned and 9.1% as dark-skinned. Despite also representing the majority of the Brazilian workforce (54.9%), Black people are noticeably underrepresented in managerial positions (29.9%). They earn much less than their White counterparts, have a greater probability of dwelling in poverty, have less schooling, are disproportionately subjected to police violence, and have higher rates of incarceration (IBGE, 2018).

Many other countries also have records of systemic racism and discrimination towards Black people, including the U.S.A., South Africa, and various Caribbean and Latin American nations. This problem is rooted in the legacy of slavery, which created a deeply rooted racial hierarchy that remains a primary factor in perpetuating that legacy (Nascimento, 2016). However, the post-slavery racial dynamics in Brazil, similar to those in other Latin American and Caribbean countries, have instrumentalized discrimination and inequality in a unique manner. Whereas the U.S.A. and South Africa had statutory regulations of segregation and disenfranchisement, Brazil's racial class system is more fluid, being primarily based on physical appearance, social status, and class level (Duc, 2018).

Despite recent progress, Brazil is still failing to address racial and structural discrimination. The economic and social rights of Afro-Brazilians are restricted, and they enjoy limited opportunities for exercising their civil and political rights (Amnesty International, 2022). In this context, our findings highlight our three fintechs' design and implementation of inclusive credit scoring algorithms as one avenue to reducing racial injustice in the financial system. The founders of these fintechs were motivated by their own personal experiences of economic inequality and injustice. This is exemplified by the story of Fintech founder 2A, who grew up in a low-income neighborhood and witnessed firsthand the challenges his parents faced in providing for their family amid economic uncertainty:

I had my childhood on the outskirts of São Paulo ..., it was all wilderness and very peripheral, so we were at the mercy of an uncertain future ... I saw my parents' struggles to feed six mouths. (Founder 2A)

Notwithstanding the significant obstacles encountered, the founders behind the three Black fintechs were able to achieve economic success in their respective areas of expertise. Still, they recognized the critical underrepresentation of Black individuals in the social arenas they frequented and the pervasive structural barriers based on race hampering their economic ascent.

As I climbed the ladder in this company (talking about their prior work), I saw fewer and fewer peers, especially Black peers and women. I hit the famous glass ceiling. Above my glass ceiling was a boss who was a combination of sexist, racist, and homophobic. (Founder 1A)

Injustice based on ethnic-cultural aspects of race was present in the form of discrimination in handling credit applications. This connection between the cultural and economic dimensions highlights how characteristics associated with Black individuals are imbued with preconceptions suggesting that they "possess less economic capacity."

[I asked for a loan from my banking manager]. ... He looked at me and said, "Nice." but then he looked away and said, "I can't lend you the money." [I had] no admin issues or unpaid taxes. Everything was OK That's when I started to understand racism more deeply, structural racism. That's when I came to understand in depth that a Black person has his credit denied three times or more, because it's a social issue, a racial issue. (Founder 2A)

The emergence of Black fintechs is situated within a broader context of social injustice, and the personal stories of their

founders underscore the intersectionality of the various forms of discrimination that they have experienced. These stories also serve to demonstrate that Black fintechs have emerged as an attempt to address social justice issues through entrepreneurship by those who have directly experienced these injustices. In what follows, we will explore the strategies employed by Black fintechs to address racial injustices encountered not only by themselves but also by other Black individuals.

Strategies to Decrease Racial Injustice in Financial Credit

In this subsection, we present our results according to the main concepts integrated into our initial theoretical framework: the process of sociotechnical reconfiguration, which involves the three social justice dimensions, i.e., recognition, representation, redistribution. For each dimension, we highlight key categories and strategies applied by Black fintechs to address social injustices experienced by the Black community. The data structure presented earlier guided the presentation of this part of the results:

Strategies of Recognition (the Cultural Dimension)

Black fintechs are playing an increasingly important role in promoting the recognition and valorization of Black people's individual and collective identities as active and capable members of society. Beyond providing economic opportunities, they generate an operational symbolic/cultural dimension of social justice that challenges negative stereotypes and highlights positive aspects of Black racial and cultural elements. Therefore, the name we give to the main category associated with this dimension is *race/class recognition*. The efforts of Black fintechs contribute to the fostering of Black individual self-esteem, thereby creating a sense of Black community identity.

Fostering Black individual self-esteem: The experiences of the Black fintechs' founders, as well as their desire to support other Black individuals who did not have the same opportunities they had, has led them to develop initiatives that promote Black entrepreneurship, aligned with a broader movement of strengthening Black identity:

It's one of my fights. Before, "Black" or "African" was seen as something bad, something pejorative. In all my projects, in my companies, everything has "Black" or "African" [in the name]. It's a way to reinvent yourself, signify the word, right? (Founder 1C)

This self-affirmation of Black identity came to feature prominently in discourse among Black fintech founders, playing an important role in the recovery of Black culture as a positive force:

There was a strengthening of a Black identity in Brazil. This strengthening of Black identity goes through a political discourse of self-affirmation. (Expert 1)

This positive identity recognition exerts an inner effect on the members of the Black community, generating greater self-esteem. As our interviews indicated, Black people's participation and performance in the financial system hinges on their self-confidence. The protagonist role taken by Black people in the studied fintechs challenges the negative stereotypes and devaluating images often imposed by the dominant culture. The ability of Black people to recognize themselves as capable and deserving of equal access to economic resources looms large in promoting the financial inclusion of these groups:

When they see a Black guy, a Black woman talking about finance, talking about banking, talking about the future, it changes the mindset of our audience. (Founder 2A)

Creating Black community identity: The goal of creating a sense of group identity among Black people is a common thread running through the initiatives of our three fintechs. By identifying Black people as a group, they acknowledge the shared injustices they face and the spaces where Black people are underrepresented. This provides a foundation for working together around a shared positive identity agenda. By creating a group identity, they produce a discourse that resonates with other Black individuals.

We [talking about Fintech C] need to have a specific speech of Black thought for Black. It is a bank of peripheral people thinking about the periphery, thinking about the Black community. We [talking about the Black community] never had that. It is a bank that proposes to bring together more and more Black and peripheral people. (Service-Provider 1C)

In addition to promoting recognition of race through affirmation and valorization of ethnic-racial differences, Black fintechs are also working to challenge the institutionalized values inscribed in the financial system that obstruct the fair participation of marginalized communities in the economic sphere. Our research identified two core strategies employed by Black fintechs to achieve this goal: to resignify race as class and to resignify peripherals as market potential.

Reframing race as class: Black fintechs intentionally assert Black identity as a symbol of economically marginalized people, framing it as representative of those who belong to a lower economic class.

[The Fintech A] is an institution where the protagonism is Black and when we look at the market, we are saying that our focus is to serve a C, D, and E class ... But this population is almost 87%, 90% Black. (Manager 1A)

The three fintechs view this lower economic class, which they refer to as "peripheral," as being primarily defined, not by their ethnic and cultural aspects, but by their common marginalized economic status. They recognize that race is closely intertwined with class and that Black individuals suffer disproportionately from both symbolic and economic injustices. Therefore, for Black fintechs, race becomes a signifier of class, with Black identity being used to represent those who are economically discriminated against, including other marginalized groups, regardless of their ethnicity.

Race is the experienced class. You have an articulation of the social dynamics in which race becomes a signifier of class. But, obviously, when we talk about the Black population, we are talking about how being Black means being poor. Race works like a stigma; it always brings you down. (Expert 1)

Promoting peripherals as market potential: Once they portray Black individuals as representative of peripherals, Black fintechs attempt to transform the way that poor communities are represented in the economy. They advocate a shift in the way peripheral communities are perceived: to view them as untapped potential markets rather than poor populations in need of aid.

This is our aim ... being acquired by a large industry or convincing that the venture capital industry was wrong in only investing in the [rich] and begin to understand that Brazil, in fact, is periphery and that is where development is actually born and needs to be scaled to change the trajectory of the country. (Founder 1B)

This repositioning serves to construct an image of the Black community as active and capable consumers and producers who are equally able to participate in market structures, despite the historical and systemic barriers they have faced. Seeking to dismantle a longstanding association between Black individuals and membership in a marginalized group, Black fintechs highlight the strong market potential of peripherals who, consequently, deserve greater access to economic resources. The Black fintechs' basic thrust is towards the empowerment of these communities and the redistribution of financial resources.

Today there are 35 million [of Black individuals in Brazil]. More than 10% of this public does not have a bank account, and ... the other 11% had an account but did not use the account because there was no relationship, and ... they moved 347 billion out of [the] system. (Founder 2A)

Strategies of Representation (the Political Dimension)

The interdependency of recognition and representation, as proposed by Fraser's theory, became evident as our analysis evolved. The quest for greater representation emerges from the struggle for recognition—and vice versa. We named the main category associated with this political dimension *Black people protagonism*. We identified two core strategies under this category: increasing Black protagonism within fintechs and increasing Black protagonism in the financial ecosystem.

Increasing Black protagonism within fintechs: The executive teams of the three Black fintechs are primarily Black in composition. Black individuals are well represented at various levels of the fintech's organizational structure, as reflected by the comments of Founder 1A, who described the startups as being characterized by "Black-protagonism"—diverse in ethnicity and gender but with Black individuals in key leadership positions.

[Fintech A] is a Black-protagonism company. So, all our founders are Black Obviously, we have employees who are not Black either. We do have a recruitment look into minority groups, women, people from the periphery. [In our company, there are] people of all ages, all colors, and all genders. (Founder 1A)

In our three cases, Black individuals play a central role in key decision-making processes. These individuals share the same racial and socioeconomic background as their target audience, giving them a unique perspective and grasp of their needs and making them well-positioned to effectively represent these communities within the fintech industry.

The journey was basically bringing people from the periphery (slum), our friends and relatives, to code, to be CFOs, to take positions. ... We also set up an advisory board with directors who also came from the periphery. (Founder 1B)

Increasing Black protagonism in the financial ecosystem: Our findings indicate that although Black individuals still face significant underrepresentation in the broader financial ecosystem, Black fintechs are actively working to rectify this

disparity. Our interviewees reported that this lack of representation has resulted in significant barriers to racial equity, limiting Black influence in economic matters and hindering the economic growth potential of their communities.

[About] strengthening the Black community, about what really generates access, power, and market influence. ... [Is] our money, the community's money, which always went out of the community to people who didn't return that money or its benefits to the community. ... We had to set up a financial structure that can meet the demands of this community, the Black community. (Founder 1C)

These fintechs' founders have established their "own banking" solutions as a means of addressing the inadequate representation of Black individuals within the financial mainstream.

We are the first digital [financial institution] that has Black people as protagonists. ... Now, we have our own bank. A bank that will represent ... 57% of the Brazilian population, which is Black and peripheral. (Founder 2A)

In the name of the Black community, they engage with venture capitalists to secure more resources for Black-owned businesses. They also serve as Black people's representatives of the fintech industry in regulatory matters.

We are the only Black points in the ... several fintechs that participate in regulatory issues, and we are the only one that has raised the flag of diversity to encourage within this entity the support of startups and fintechs led by Black people. ... Our responsibility was to go beyond the service we provide. (Founder 2A)

Strategies of Redistribution (the Economic Dimension)

Although the range of financial services provided by our three fintechs to Black communities is broad—from digital wallet to microinsurance—we have focused specifically on financial credit granting services because they are directly relevant to our research question. Our analysis was guided by the sensitizing dimension of redistribution, which allowed us to identify *credit access* as one key category. Credit access plays a crucial role in addressing financial injustice by enabling individuals and businesses to invest in their future and improve their economic well-being. This process is instrumentalized through four strategies: *running forward-looking analysis, testing affirmative variables, building contextually tailored algorithms, and integrating racial bias awareness.*

Running forward-looking analyses: The three Black fintechs acknowledge the limitations of relying solely on conventional backward-looking credit analysis, which focuses primarily on an individual's past financial data, such as previous loan payments and unpaid debts. This type of analysis, usually conducted by third-party credit agencies (bureaus), serves as a reference, but the Black fintechs understand that those techniques alone are not sufficient to accurately evaluate a Black individual's payment capacity.

These [credit] bureaus have an exclusionary mechanism. When you look at the eight items that the bureaus look at, one of the first items is zip code. This is a factor that makes this entrepreneur not have access to credit. (Founder 1A)

We are not concerned with your past, but with your future. (Founder 1B)

In the context of historical economic inequalities, assessing an individual's ability to repay a loan based solely on their financial history can deprive Black individuals of future opportunities. Backward-looking analysis relies on past financial data and does not consider other factors, such as future earning potential, that could influence a person's ability to repay a loan in the future. As a result, Black fintechs are confronting a paradoxical challenge. They must determine an accurate method for assessing the future payment potential of Black individuals without relying on variables reflecting their previous payment capacity. This requirement is crucial to ensure that the algorithms used in the credit scoring analysis are not influenced by the economic disadvantages in which Black individuals are entrenched.

[The Black fintechs] are talking about a potential market, the algorithm is looking backwards, they are looking forward. What elements go into the production of the algorithm that can provide predictability to capture this potential? ... If you only select elements from the past it won't change. (Expert 1)

To overcome these limitations, Fintech A is exploring new ways to assess creditworthiness with forward-looking analysis—a more proactive approach that uses various predictive tools and models to assess future creditworthiness. This approach considers both the borrower's historical financial data and, weighted more heavily, alternative data that could potentially inform about their future financial performance.

We are proposing to make a more prescriptive model, ... we will try to understand the future behavior that our client will have in two or three months and how much credit we can offer that does not compromise. (Developer 1A)

This same approach is followed by Fintechs B and C, which constantly strive to include alternative variables and obtain new data that might attest to the Black individual's future payment capacity without basing it on their past economic and material situation.

Testing affirmative variables: Fintech C is in the early development stages of its credit scoring algorithm but has already integrated a "social score" analysis aimed at evaluating social vulnerability for the purpose of promoting the equitable distribution of credit and donation grants. The social score considers "affirmative variables" in its analysis, which comprise both cultural (e.g., Black race) and economic (e.g., participation in income transfer programs) data that are associated with marginalized groups and can be positively weighted in credit analysis. Therefore, the social score imprints a positive bias for Black people and other marginalized groups:

[Our model has a] higher score for being Black. It's a social issue, we can't have a social score with the structural racism that exists, with access for Black people being more difficult. Occupying spaces is more difficult. So, it wouldn't make sense to have a social score balancing the issue of race. (Founder 1C)

The integration of social scoring into conventional credit scoring models has enabled Fintech C to conduct a more inclusive and holistic analysis for financial credit and donation granting. Rather than relying solely on a forward-looking analysis of a Black individual's payment potential, Fintech C's experimentation incorporates a variety of economic and social data to provide a credit analysis that goes beyond the individual's creditworthiness and also takes into account historical and systemic inequalities faced by the Black community.

Sometimes the negative credit was due to mismanagement, a difficult moment and a need. This is all part of our score. ... But [additionally] there are some parameters that we cannot change. There are some parameters that are already part of the social score. (Founder 1C)

Building contextually tailored algorithms: In our study, we observed that Fintech B had a more advanced credit scoring algorithm than the other two Black fintechs, as most of it had been developed in-house, allowing them to exercise control over the entire credit granting process from application to disbursement. To grant credit to marginalized individuals who would be denied based on traditional backward-looking analysis, Fintech B not only runs forward-looking analysis but has also built tailored algorithms for different social groups. For example, to grant credit to women, Fintech B developed a

new algorithm to evaluate the payment potential of single mothers. This algorithm incorporated contextualized factors such as alimony payments, details about family structure, and the type of school their children attend.

To increase credit for Black women in particular, we were unaware of the alimony variable as a score generator. ... This made us create an algorithm only to evaluate single mothers, under the premise of alimony as an additional income. (Founder 1B)

Although tailored credit analysis plays an important role in Fintech B's operation, it is not their major mode of operation. The company relies primarily on conventional credit scoring models to assess the majority of its clients. Personalized and granular credit granting analysis is only conducted for those individuals who have been unable to access credit through conventional methods, such as those with a prior history of unpaid debts. The company's credit analysis is specifically designed to address the particular context of minority groups, considering alternative questions and variables suited to each group's profile.

For each one [credit analysis] there is a range of variables that change as each question is inserted. But when someone has a [good] credit score [in credit bureaus] we don't even need to take [these other variables] into account. [The credit analysis with our software] starts from the third, fourth phase of the analysis. Up to the first and second [it] is only outsourced. (Founder 1B)

Integrating racial bias awareness: The unique set of variables and contextually tailored algorithms used by these Black fintechs to evaluate specific marginalized groups are derived from the extensive knowledge and racial bias awareness possessed by their own developers and managers. This firsthand understanding of their customers' social and economic context provides deeper insight into the needs and behaviors of these communities. The integration of racial bias awareness with algorithmic intelligence is a key component in developing inclusive premises for credit granting in Black and marginalized communities.

For example, Fintech B's founders' and managers' own personal experiences and awareness of racial discrimination were critical in developing innovative and inclusive credit evaluation practices for slum residents. They believe that property deed records are a better predictor of future payment capacity in slum communities than traditional indicators, such as ZIP code, reflecting their deep understanding of the social and cultural context of this particular population.

If you manage to obtain a deed record in the slum [favela] ... if you achieve this feat, you have already demonstrated a lot of things. So, in the algorithm, we don't even look at the rest when a resident presents this type of data. (Founder 1B)

The premises involving racial bias awareness are subjected to small-scale experimental testing, which is conducted manually. They are integrated into the algorithm only when they have been proven to be effective and when scaling is required.

So, we don't need to customize the algorithm as much. We did it manually and we tested it manually. (Founder 1B)

Although all three Black fintechs have engaged in manual human-driven experimentation, Fintech B demonstrated a more systematic approach. This involved leveraging the company's financial resources and a portfolio of small testing clients to refine the credit scoring analysis. Once incorporated into an algorithm, it is continually evaluated for accuracy and inclusiveness. Human insight and judgment remain crucial in this ongoing evaluation in order, for example, to identify why the algorithm included someone who defaulted on payments or excluded someone who should have been included. This iterative process serves to improve the algorithm's predictive power and ensure fairness and inclusiveness in credit assessments.

We analyze the [credit] model's decision. The yes or no of the model was due to what? Then we take samples of the results ... and we analyze those samples in the light of our experience of human contact with people to really understand if the model was right or if it was wrong. (Founder 2B)

Presenting a Theoretical Framework: The Process of Affirmative Experimentation

Our findings reveal the process of sociotechnical reconfiguration to be one of continuous *affirmative experimentation* aimed at creating digital solutions that promote social justice and address racial discrimination in the financial sector. This process involves an interactive approach of learning and development, undertaken by individuals committed to combating race- and class-based financial exclusion, with a focus on creating credit scoring algorithms that are inclusive and empower the Black community to participate as equal members of the financial system. This final theoretical framework is presented in Figure 3 and synthesizes the concept of affirmative experimentation, drawing on the results of our case studies of three Black fintechs.

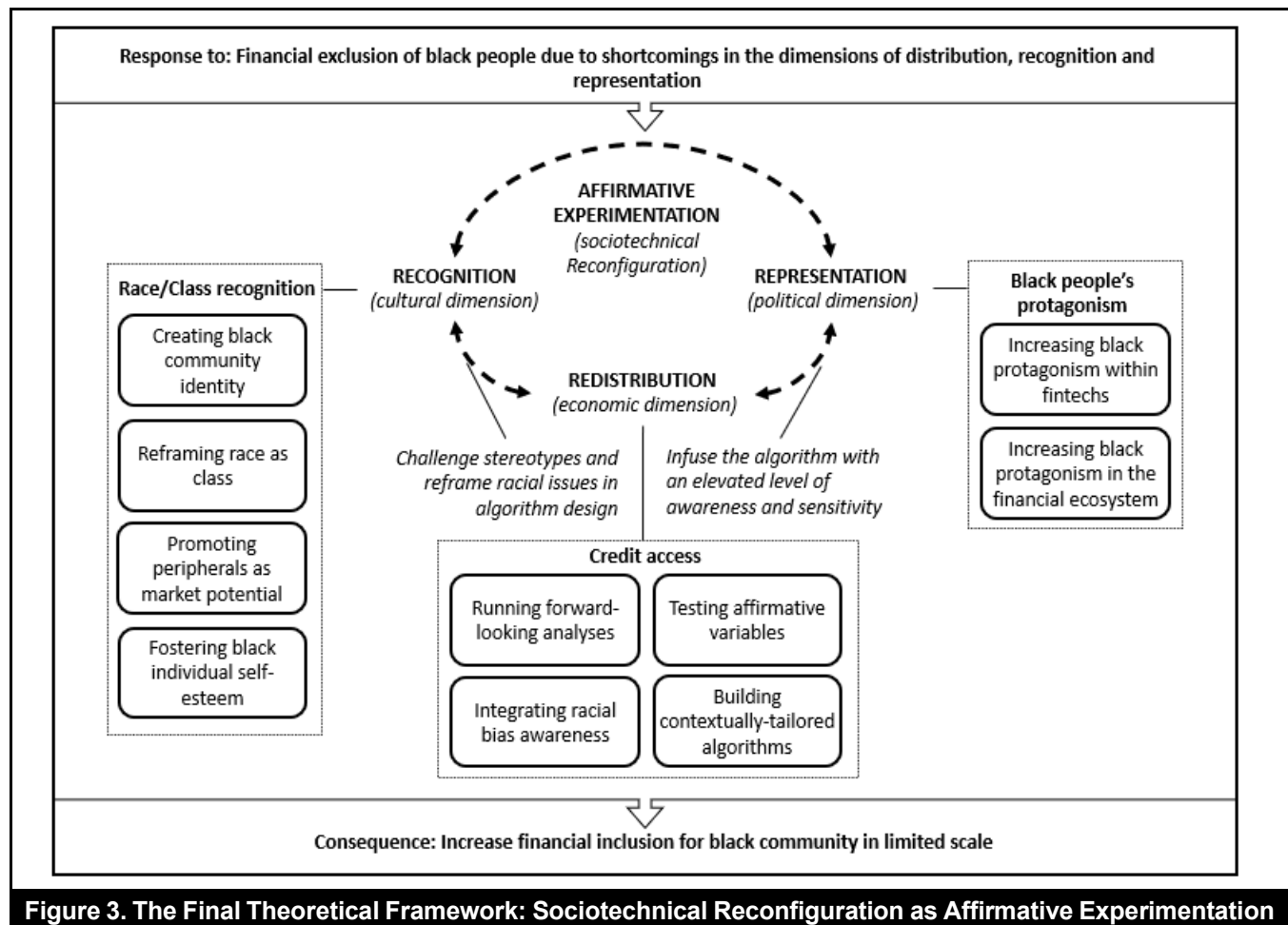


Figure 3. The Final Theoretical Framework: Sociotechnical Reconfiguration as Affirmative Experimentation

Our framework illustrates that Black fintechs operate across the three dimensions of social justice—recognition, representation, and redistribution—in an ongoing sociotechnical reconfiguration. This process involves an interplay between technical elements and social values, with continuous exchange between them in a dynamic and evolving process towards social justice. By focusing on the strategies employed by Black fintechs, the framework provides insights into how they are leveraging affirmative experimentation to address longstanding racial disparities in the financial industry. The framework also illuminates the triggers and consequences of affirmative experimentation.

First, our findings demonstrate that Black fintechs in Brazil have emerged as a response to systemic social injustices, as manifested, in particular, in the financial exclusion of Black people due to shortcomings in the dimensions of distribution, recognition, and representation. By incorporating their own lived experiences, Black fintechs are uniquely positioned to challenge implicit racial bias in the development of credit scoring algorithms (Martin, 2019).

Second, our findings reveal how Black fintechs act by developing strategies in the three dimensions of social justice. Race/class recognition strategies actively challenge prevailing negative stereotypes and strive to reframe racial issues through algorithm design. These strategies not only foster a collective awareness of systemic injustices but also serve to recast racial disparities within a broader framework of class inequality. Ultimately, such initiatives serve to empower marginalized communities by addressing the multiple layers of social inequity they face. Race/class recognition acknowledges the pervasive influence of racial structures within institutional contexts, as highlighted by Ray (2019). This involves a proactive effort to redefine Black identity symbolically, countering longstanding ethno-racial prejudices that permeate cultural understanding. Importantly, our theoretical framework illuminates that the racial bias encountered by fintech companies is twofold: it is both cultural and institutional in nature. Hence, racial biases are not just confined to individual attitudes or societal beliefs but are deeply embedded within economic structures that govern access to financial resources, thereby exacerbating

disparities in financial inclusion. Our findings show that recognition is associated with scoring algorithms when they are designed to embody the concept of socially excluded people valuing their own identities, with a view towards transforming the symbolic order embedded in the exclusionary reality. This means that once minorities are represented within the development team, this group must be able to identify which variables used in the algorithm can leverage greater exclusivity or inclusivity.

In terms of representation, the strategies of Black fintechs focus on increasing the protagonism of Black individuals, both within their organizations and in the larger financial industry. Despite the underrepresentation of Black individuals in the financial sector (Florant et al., 2020; Reynolds, 2020), our findings show that Black fintechs have increased the representation of Black individuals in decision-making positions, endowing these organizations with a unique racial perspective. Our framework highlights the significance of this protagonism in meeting the financial needs of Black communities, as it allows the design of credit scoring systems with heightened awareness of racial bias. Therefore, representation is connected to the design of scoring algorithms through the concept of excluded people themselves taking the lead in the process aimed at their inclusion—meaning that minorities must be included in the algorithm development team. The distinguishing clout of Black fintechs, in line with the demands of the Black money movement (Soares, 2021), is their ability to promote racial fairness due to the presence of Black individuals in key decision-making positions. These individuals possess firsthand experience of racism and discrimination, and their backgrounds often mirror the contexts in which their clients live. As a result, they bring an elevated level of awareness and sensitivity to credit scoring algorithms, particularly with regard to the far-reaching impacts of structural racism.

Redistribution strategies seek to provide credit access to marginalized communities. Our study identifies four strategies that emphasize the pivotal role of inclusive credit scoring algorithms and the significance of human interactions in their development and mediation. Additionally, our framework suggests that assessing credit scoring analysis based solely on one's ability to repay can have limited inclusion impacts. The proposed solution entails introducing affirmative variables that assign positive values to socioeconomic characteristics associated with marginalized groups. This leads to a positively "biased" algorithm that favors the inclusion of marginalized individuals. Thus, by means of a higher score, it not only attempts to mitigate racial bias in credit scoring algorithms (Vilarino & Vicente, 2020; Florant et al. 2020) but also seeks to favor those suffering from social injustice.

Furthermore, our findings show that the complexity and diversity of the Black community require detailed analysis closely linked to the specific context and situations of its members, which implies an acknowledgment that they experience various types of injustices beyond race. Developing contextually tailored algorithms for credit redistribution takes into account the social and economic context in which an individual operates, leading to a more comprehensive assessment of creditworthiness. Likewise, our findings highlight the importance of considering human awareness and cultural sensitivity when developing credit scoring algorithms. By leveraging their personal experiences and expertise, Black fintechs can introduce awareness of racial bias into algorithm design and credit scoring analysis. This is in line with the literature emphasizing the role of human subjectivity in the design and mediation of credit scoring algorithmic decision-making (van den Broek et al., 2021; van der Aalst, 2021; Lehr & Ohm, 2017; Yarger et al., 2020). However, Black fintechs go beyond simply integrating humans into the process of designing inclusive algorithms by introducing racial bias awareness that can lead to more inclusive and accurate assessments of creditworthiness.

Finally, our framework shows that as a consequence of affirmative experimentation, Black fintechs have the potential to increase financial inclusion for the Black community, albeit on a limited scale. Their capacity for redistribution is limited by various factors, including the lack of adequate regulatory and public policies (Baradaran, 2018), and restricted access to human and social capital (Barki et al., 2021). One significant limiting factor is their restricted access to the financial resources needed to create technology-intensive credit score analysis and credit funds to scale up their operations and develop their own credit scoring tools—an unjust restriction shared by the individuals they seek to help. Moreover, the challenges faced by Black fintechs go beyond financial access and include social and systemic barriers that inhibit participation by Black individuals in the fintech ecosystem. Yet despite these challenges, Black fintechs have demonstrated the ability to raise awareness of the needs and challenges facing Black communities, promote participation in the fintech industry, and open a path towards financial inclusion for the Black community.

Implications of our Study: Do Black Fintechs Matter?

To better understand if Black fintechs matter to social justice, we undertook a qualitative study of how Black fintechs design and use inclusive algorithms to decrease persistent social injustice involving racial financial inclusion. In this section, we discuss the main implications of our study with regard to the extant literature.

Our literature review indicated that promoting financial inclusion to Black communities requires a coordinated effort on many social fronts of society because it cannot be achieved solely through the usual actions towards improved financial inclusion for low-income people in general (e.g., Sanchez-Moyano & Shrimali, 2021). Second, it is necessary to ensure that Black communities take a leading role in the fintech universe to avoid algorithmic bias based on the use of variables that reflect structural inequalities (e.g., Reynolds, 2020). Third, combating racial financial exclusion needs to be based on a combination of human and machine interactions to improve both the development of unbiased algorithms and the decisions taken based on credit scoring systems (e.g., van den Broek et al., 2021).

Although important in pointing out the problem of racial financial exclusion, these lessons do not offer a clear path to formulating inclusive algorithms that could improve the Black community's access to credit. Nonetheless, our findings suggest that Black fintechs *are* making a sound contribution towards social justice in Brazil. However, given the structural dimension that excludes Black minorities, we also found that the development of inclusive algorithms with the potential to overcome that exclusion is a gradual process that involves testing affirmative variables and integrating racial bias awareness in the development of credit scoring tools.

Concerning the implications of our conceptual framework, we contend that combining the concept of sociotechnical configuration, part of the South American research tradition known as *tecnologia social* (Dagnino, 2019), with the three seminal concepts of Fraser's (2005) theory of social justice—recognition, representation, and redistribution—constitutes a significant theoretical contribution to the IS field.

Our review of literature drawing on Fraser's ideas found some studies simply mentioning her seminal work (e.g., Levy et al., 2020) but very few studies directly applying her concepts, Marjanovic et al. (2022) and Bek and Aygün (2016) being among the few exceptions. The latter authors suggest that further studies on inequalities in digital platforms should adopt Fraser's social justice approach. As our own analysis evolved, our enthusiasm warmed as we discovered how well Fraser's theory of social justice sheds light on the tensions and struggles experienced by Black people operating in the fintech ecosystem—underscoring the priority of overcoming the restrictive effects of racial discrimination. Consideration of the relationship between affirmative and transformative action deserves special attention.

Fraser (2005) is categorical in affirming that true progress towards social justice depends on transformative action, meaning profound or radical changes in the structural sources of social injustice. Fraser acknowledges that affirmative

action does play an important role in helping to produce a fairer social world, but she reminds us that it is not enough. Our literature review on race-based financial exclusion shows that the complexity of this problem cannot be addressed exclusively through market approaches, such as financial institutions exclusively dedicated to serving Black communities (Kashian et al., 2014; Chiong et al., 2018). Our review also shows that increasing direct minority participation in financial institutions is a necessary part of the solution (Florant et al., 2020; Eyal-Cohen, 2021).

Highlighting the relevance of active participation by Black communities in the quest for social justice in the financial system, our investigation of the Brazilian Black fintechs demonstrates the long and winding path traced by the development and use of inclusive algorithms. We view the development and use of such algorithms by Black fintechs as an example of affirmative experimentation. We conceive affirmative experimentation to be a process of sociotechnical reconfiguration—involving the political, symbolic, and economic struggle for the recognition and participation of Black people as equal members of the market mechanisms—as well as the pursuit of changes towards equalizing financial resource redistribution (Figure 3).

Black fintechs exemplify recent advances in Black communities' recognition and representation in the fintech universe, and they hold the potential to catalyze the economic redistribution sought by this minority group. Recognition is progressing on two main fronts: community identity recognition and market status recognition. These struggles for recognition have been accompanied by advances in representation: Black fintechs are owned and managed by the very same Black people that are fighting for recognition. However, an appraisal of redistribution can only be expected to take place over the long term. We explain this imbalance among Fraser's three categories in terms of the complex interplay between micro-meso (affirmative) and macro (transformative) changes needed to achieve social justice in the financial system. Affirmative experimentation lies at the heart of our Black fintechs' efforts toward racial justice in Brazil, based on small incremental advances in the scope and scale of the Black community's financial inclusion. Significantly, this gradual incremental advance is similar to that achieved by Black-owned banks in the U.S.A. (Neal & Walsh, 2020).

In line with previous findings in the literature (Robb & Robinson 2018), Black fintechs use forward-looking analysis to evaluate creditworthiness while also availing themselves of other forms of analysis. Our studied cases demonstrate Black fintechs' struggle to actualize their own view of credit scoring by testing affirmative variables and integrating racial bias awareness into their algorithms. For example, Fintech A has

undertaken the development of its own credit scoring algorithm, planning to bring in its original approach to redistribution at a later time. Fintech C is developing an innovative algorithm that incorporates a “social score” into its creditworthiness assessment by including affirmative variables that reflect the realities of marginalized communities, thus injecting a positive bias for Black and other marginalized individuals into credit redistribution. The struggle is even more evident when dealing with other relevant actors in the fintech ecosystem. Through ongoing experimentation and the refinement of its credit assessment process, Fintech B seeks to demonstrate to investors that the inclusion of marginalized populations can be profitable, concomitantly reinforcing its own political positioning in the quest for financial justice.

It should be noted that the struggle surrounding such affirmative experimentation takes place within the institutionalized barriers of the financial ecosystem and the established market rules that constrain Black fintechs. As the credit industry operates under a risk-return logic, Black fintechs are competing for access to funds with more traditional players who use conventional credit scoring tools. Faced with economic class structures, the Black fintechs’ challenge is to find the right balance for Black people’s inclusion—one that maximizes actual return potential by accurately identifying individuals with such potential. Affirmative experimentation means creating new ways to overcome these institutionalized barriers while recognizing that significantly scaled inclusion will happen only in the long term.

Regarding the design and use of algorithms, Black fintechs constantly experiment with new variables by using a forward-looking approach to reconfigure the traditional backward-looking approach to credit scoring (Robb & Robinson, 2018). Thus, Black fintechs adopt improvements based on human analysis combined with automated analysis, yielding a richer perspective for the credit scoring process (van der Aalst, 2021) that produces more inclusive results. Over time, the knowledge acquired by Black fintechs through this human-machine interaction (van den Broek et al., 2021) can be incorporated into new and more powerful inclusive algorithms. This serves to challenge the established market logic and generate new ways to evaluate poor Black individuals who are usually excluded from the credit instruments that promote social and economic development.

Overall, we found that a complex tension exists within Black fintechs, reflecting the fragile balance between their goal of increasing the scale and scope of financial redistribution and their pursuit of resources—financial and technical—and of enhanced reputation. The most important constraint is that imposed by the risk-return logic of investment funds (the

ecosystem), for whom the struggle against class or racial injustice is not an issue. Here we concur with Fraser’s view that transformative action should imply the structural social changes necessary to eliminate injustice at its societal roots.

Regarding the analytical generalization of our results, they are consistent with previous research on algorithmic fairness showing the need for algorithm designers to be aware of their own biases in their particular situated context. Such context may include the prediction of costs associated with population-based health care systems (Obermeyer et al., 2019), eligibility for social services (Martin, 2019), and probability assessment for criminal recidivism (Chouldechova, 2017). Therefore, we consider it plausible that our findings will be applicable not only to financial systems and credit scoring algorithms but might also encompass a wider range of industries where algorithms have proven decisive in predicting minorities’ future behavior. In this regard, we highlight our case’s three-dimensional potential.

First, we consider the economic, cultural, and political sociotechnical reconfiguration process uncovered in our research to hold relevance for addressing a variety of digital bias in other types of algorithms. We understand that our discussion of credit scoring algorithms is instrumental, and our cases stress examples of particular kinds of algorithmic variables. Nevertheless, the categories that emerged from our study, e.g., “testing affirmative variables,” can be extrapolated to any algorithmic system wherein the decision-making process hinges on anticipating the future behavior of particular social groups. For instance, the same principles might be applied to systems like those for healthcare predictive analytics, criminal justice risk assessment, and job candidate screening.

Second, while Brazil has unique socioracial characteristics, our results might be extrapolated to other economic-cultural contexts that share historical connections to slavery, where racial miscegenation persists simultaneously with discrimination against Black people. The underrepresentation of Black individuals in the fintech ecosystem is not a localized phenomenon (Brummer, 2020). As evidenced by statistics, racism is also present in that Black applicants experience significantly higher rejection rates in their loan applications than White applicants (Mikhitarian, 2018). This discrepancy is further elucidated by the emergence of Black-owned banks (Kashian et al., 2014), a phenomenon similar to the rise of Black fintechs in Brazil. Moreover, the advent of the Black Money movement, originating in the U.S., has significantly shaped the landscape of Brazilian Black fintechs.

Third, we see opportunity to extrapolate our results to the overall dimension of minorities. While our work focuses on the Black community, we have seen similar initiatives take

shape within other minority groups—particularly, women and LGBTQIA+. Nonetheless, we acknowledge that this extrapolation needs to be finely calibrated in order to take in the misogynistic and homophobic elements faced by these groups, whereas the dimensions of race and class might not necessarily be present. Hence, the relevance of our findings is not confined to a single minority group. Rather, the impact of sociotechnical reconfiguration may extend to other marginalized groups that encounter similar discrimination due to their societal standing, such as immigrants, refugees (Molnar, 2019), and women (Hall & Ellis, 2023).

Thus, in a world where algorithms have become more pervasive in decision-making processes affecting human lives, there is evident need to prevent the unintended negative consequences and widespread discrimination that frequently occur when decision-making agency is transferred from humans to algorithms. In this sense, our findings suggest that regardless of the domain in which the tool is applied, marginalized people need to shape the design of the algorithm in ways that affirm their value and identity.

Final Considerations

We contribute to IS research theory concerning social justice by applying the concept of sociotechnical reconfiguration—involving Fraser’s concepts of recognition, representation, and redistribution—in an analysis of three Brazilian Black fintechs. Additionally, we highlight how Black fintechs might catalyze changes in the underlying social structures of the fintech market and how such alternative logic opens different design paths for using and creating credit scoring algorithms through a process we term affirmative experimentation. We see our research as contributing to the conceptualization of societal change in terms of shifts in the underlying values and assumptions of fintechs—a mechanism that Faik et al. (2020) call translating. In our case, we see Black fintechs creating new patterns to develop inclusive algorithms, adopting an affirmative experimentation logic that differs from the dominant practice of implementing algorithms that reinforce racial discrimination.

Finally, we also offer important managerial contributions by providing a better understanding of this affirmatively experimental approach to designing inclusive algorithms. Our results indicate that sociotechnical reconfiguration is involved in the construction of socially inclusive algorithms, culminating in affirmative experimentation that promotes minorities’ inclusion in complex social processes and, in the long term, social justice. Our research findings highlight the importance of sociotechnical reconfiguration in reducing biases within financial algorithms. To achieve this goal, financial institutions should embrace a participatory approach

in which the socially excluded individuals take a leading role in the process in a way that recognizes and values their identities. Moreover, algorithms should incorporate a combination of human and automated analysis to effectively ward off exclusion. These are the crucial elements for the development of inclusive algorithms that promote a fair and equitable financial system—an evolution important not only to the excluded minorities but also to the whole economy (Sanchez-Moyano & Shrimali, 2021).

Nonetheless, our study has certain limitations. A methodological limitation involves the number of interviews. Initially, we planned to conduct a greater number of interviews with clients and technical staff. However, issues related to confidentiality between the fintechs and their clients and service providers limited our ability to obtain interviews with these key stakeholders. Despite these constraints, we were still able to gather valuable insights from all managerial and technical leaders involved in the algorithm development process. In terms of the development team itself, the majority are contracted from external third-party companies and we were unable to interview them. Interviews also could not be arranged with leaders involved in the Black money movement, although we reached out to them. Additionally, our data is limited to the cultural and regulatory context of Brazil. As such, although our findings may be analytically generalized to other similar contexts, the specificities of racism and financial regulatory issues in different regions may limit the direct transferability of our results. It is also important to note that our study lacks a discussion of the local regulatory environment, which has exercised an undoubted influence on the development of Black fintechs in Brazil.

During our research, we organized one workshop to present the preliminary results of our research to members of Brazilian Black movements. Their main concern was two “silences”: the absence of any Black members of our research team and the absence of Black Latin American thinkers in our theoretical framework. These weaknesses could have been overcome by integrating the work of important Black activists and theoreticians like Abdias do Nascimento, Djamila Ribeiro, Silvio de Almeida, Sueli Carneiro, Lelia Gonzalez, Conceição Evaristo, and Milton Santos in our theoretical background and/or methodological approach.

We see this study as the very first step of a thorough and relevant research program. First, future studies on Black fintechs could benefit from incorporating local Black scholars and intellectuals into their conceptual frameworks to take into account the cultural and economic nuances of racism in each country. Furthermore, future studies might also focus on the potential social justice contribution of Black fintechs in other countries where Black communities also suffer from structural discrimination in different societal spheres.

Additionally, as the founders of Black fintechs can be considered social entrepreneurs (Kimmitt & Muñoz, 2018), future studies could further explore the role of Black social entrepreneurs. This could include their personal experiences with social inequalities as catalysts for the development of alternative and more inclusive logics, as well as their interactions with the wider fintech and social ecosystems in which they operate. Examining the context of structural racism (Nascimento, 2016) within financial systems would provide valuable insights into how Black fintechs might leverage those ecosystems in advancing social justice and financial inclusion and would enrich understanding of the actual process of developing tailored digital solutions. A deeper understanding of Black fintechs could also be achieved by further investigation of the regulatory environment and by carrying out comparative studies that include Brazilian Black fintechs alongside North American or African Black fintechs.

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